

Andrew Pham

Toronto, ON, Canada | andrew.pham@example.com | linkedin.com/in/andrewphamml | github.com/andrewphamml

SUMMARY

Computer Science student specializing in machine learning systems, research engineering, and high-performance software. Experienced with Python, C++, PyTorch, JAX, TensorFlow, distributed GPU training, computer vision, Transformers, and production ML infrastructure. Built large-scale model training pipelines, autonomous-driving perception prototypes, GPU inference systems, and backend services.

Seeking Machine Learning, Research Engineering, ML Systems, or Software Engineering internships.

EDUCATION

University of Toronto

Bachelor of Science in Computer Science, Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Robotics, Distributed Systems, Computer Systems, Algorithms and Data Structures, Probability and Statistics

EXPERIENCE

Machine Learning Research Intern

May 2026 – Aug 2026

Vector Intelligence Lab

Toronto, ON

- Built distributed transformer training pipelines using PyTorch Distributed, DDP, and FSDP across 8 NVIDIA GPUs.
- Trained and evaluated autoregressive Transformer models on large text and multimodal datasets.
- Benchmarked PyTorch, JAX, and TensorFlow implementations of representative ML workloads.
- Implemented mixed-precision training, gradient accumulation, checkpointing, and distributed data loading.
- Improved GPU training throughput by 2.3x through profiling and optimization.
- Developed reproducible research pipelines using Weights & Biases and automated experiment configuration.
- Collaborated with researchers to implement and evaluate new model architectures.

Machine Learning Engineering Intern

May 2025 – Aug 2025

Autonomy Systems Canada

Toronto, ON

- Developed computer-vision components for an autonomous-driving perception prototype.
- Built multi-camera pipelines for vehicle and pedestrian detection using PyTorch and OpenCV.
- Processed large image and video datasets for model training and evaluation.
- Implemented evaluation tools for precision, recall, latency, and failure-case analysis.
- Optimized GPU inference pipelines, improving throughput by approximately 2.7x.
- Wrote performance-critical preprocessing and geometry components in C++.
- Built Python APIs for running experiments and inspecting model outputs.

Software Engineering Intern

May 2024 – Aug 2024

Northstar Software

Toronto, ON

- Developed backend services using Python and FastAPI.
- Designed REST APIs and PostgreSQL schemas for production applications.
- Implemented authentication, validation, logging, and asynchronous background jobs.
- Added automated unit and integration tests using pytest.
- Containerized services using Docker and contributed to CI pipelines.
- Participated in design reviews, code reviews, and production debugging.

PROJECTS

Distributed LLM Training System | Python, PyTorch, FSDP, CUDA

- Implemented an autoregressive Transformer training system.
- Scaled training from a single GPU to 8 GPUs using DDP and FSDP.
- Added mixed precision, gradient accumulation, distributed data loading, and checkpoint recovery.
- Profiled GPU utilization and collective communication overhead.
- Compared throughput and memory usage across single-GPU, DDP, and FSDP configurations.

Autonomous Driving Perception System | Python, PyTorch, OpenCV, C++

- Built a perception prototype using synchronized multi-camera video.
- Implemented vehicle, pedestrian, and road-object detection.
- Developed preprocessing, visualization, and evaluation tools.
- Investigated model failures under low light, occlusion, and motion blur.
- Optimized parts of the inference pipeline in C++.

ML Inference Service | Python, FastAPI, PyTorch, Docker

- Built a production-style inference API for GPU-hosted ML models.
- Implemented dynamic batching and asynchronous request processing.
- Added latency and throughput monitoring.
- Improved throughput by approximately 2.7x compared with the initial implementation.
- Containerized the service for reproducible deployment.

JAX Transformer Experiments | Python, JAX, Flax

- Implemented Transformer attention and feed-forward layers in JAX.
- Used JIT compilation and vectorization for accelerator execution.
- Benchmarked equivalent JAX and PyTorch workloads.
- Investigated compilation, memory, and throughput tradeoffs.

Research Experiment Platform | Python, FastAPI, PostgreSQL, Docker

- Built a small platform for launching and tracking ML experiments.
- Stored experiment configurations, metrics, artifacts, and model checkpoints.
- Added APIs for comparing runs and recovering interrupted experiments.
- Integrated Weights & Biases for experiment visualization.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, CUDA, SQL, TypeScript, Bash

Machine Learning: PyTorch, JAX, TensorFlow, Hugging Face Transformers, OpenCV, scikit-learn, NumPy, Pandas

Large-Scale ML: PyTorch Distributed, Distributed Data Parallel, Fully Sharded Data Parallel, Multi-GPU training, Mixed-precision training, Gradient accumulation, Model checkpointing, Distributed data loading

Computer Vision and Robotics: Object detection, Image segmentation, Multi-camera perception, 3D geometry, Sensor data processing, Representation learning, Multimodal models

Backend and Software Engineering: FastAPI, REST APIs, PostgreSQL, Redis, Docker, Linux, Git, GitHub Actions, pytest

ML Infrastructure: CUDA, NVIDIA Nsight, Weights & Biases, FAISS, Docker, Linux GPU environments

RESEARCH INTERESTS

- Large-scale machine learning - Foundation models - ML systems - Computer vision - Autonomous systems - Representation learning - Multimodal learning - Efficient model training and inference

ACHIEVEMENTS

- Presented distributed-training experiments at an undergraduate research symposium. - Finalist in a university machine learning competition. - Contributed to an undergraduate autonomous-systems research project. - Built multiple GPU-based ML systems outside coursework.

LANGUAGES

- English — Fluent - French — Intermediate